
The mortality burden from COVID in low-income settings: evidence from verbal autopsies in India

Received: 26 April 2025

Accepted: 5 January 2026

Published online: 22 January 2026

Cite this article as: Malani A. The mortality burden from COVID in low-income settings: evidence from verbal autopsies in India. *BMC Public Health* (2026). <https://doi.org/10.1186/s12889-026-26215-9>

Anup Malani

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

The Mortality Burden from COVID in Low-Income Settings: Evidence from Verbal Autopsies in India

Anup Malani^{1*}

¹University of Chicago, Chicago, Illinois, USA

* Corresponding author

Email: amalani@uchicago.edu (A. M.).

ARTICLE IN PRESS

Abstract

Measuring the mortality burden of SARS-CoV-2 in lower-income countries is difficult because death registries are incomplete and lack cause of death. As a result, it is difficult to determine if deaths during the COVID pandemic were directly caused by SARS-CoV-2 or another ailment. We address this problem in India, which had the second-highest number of officially reported cases. We completed WHO-compliant verbal autopsy (VA) surveys on roughly 20,000 deaths drawn from a population-representative sample of people who died between 2018 and 2021. These surveys identify the cause of death for deceased persons via surveys of their next of kin. We find that deaths attributable to SARS-CoV-2 spike in June 2020, just after India's lockdown, and in May 2021, after its second wave. During those spikes the virus is responsible for 23.3% and 35.8% of all deaths, respectively. Cardiovascular deaths also spike during the start of the pandemic. We find that the death rate rises by 81% during the pandemic, SARS-CoV-2 is directly responsible for 33% of those excess deaths, and cardiovascular disease for 23% of those deaths.

Introduction

Even as the COVID pandemic has eased, it is important to evaluate its impact. Measuring its health impact more precisely is critical for determining both the effectiveness and cost-effectiveness of policy responses, including non-pharmaceutical interventions such as lockdowns and school closures and medical interventions such as vaccines. Estimating the mortality burden of SARS-CoV-2, however, is no easy task. In higher-income countries, where there are well-developed reporting systems, there remain debates about causation ^{1,2}. In lower-income countries, where such systems are underdeveloped or missing, the challenges are greater still ³. India is a case in point.

India is among the countries that were hit the hardest by the COVID pandemic. Through 2021, it reported the second highest-number of confirmed SARS-CoV-2 cases (~44 million) and third-highest number of SARS-CoV-2 deaths (~530,000) of all countries ⁴. Even these high numbers might underestimate the virus's impact on the country. Estimates of excess deaths—the increase in all-cause deaths from 2020 to mid-2021 relative to baseline—range from 4 to 6 million through mid 2021 ⁵⁻⁹.

These excess mortality estimates are contested, however, because India has poor measures of deaths specifically attributable to SARS-CoV-2. Estimates of excess deaths are typically based on India's state Civil Registration System (CRS). While those collect data on deaths, they do not attribute a cause to deaths ¹⁰. (Non-official surveys that report deaths have similar problems ^{5,6}. Therefore, they may over-estimate deaths attributable directly to SARS-CoV-2. During the pandemic, India began to report the number of decedents who tested positive for SARS-CoV-2 or were presumed to be SARS-CoV-2 positive. These are the basis of official COVID death reports. However, these reports may underestimate SARS-CoV-2 deaths because a

small percentage of decedents were tested or diagnosed for SARS-CoV-2¹¹. India's Sample Registration System (SRS), which samples a random 1% of the population, conducts verbal autopsies to determine cause of death¹⁰. However, the SRS for 2020-2021 has not yet been completed.

To determine how much of excess mortality is attributable to SARS-CoV-2 as opposed to the pandemic overall, we obtain a representative sample of decedents in India and ascertain cause of deaths using WHO-compliant verbal autopsy surveys administered to next-of-kin. The representative sample was obtained from the Consumer Pyramids Household Survey and contained all deaths reported in that survey between 2018-2021. The verbal autopsies revealed cause-of-death encoded in 3-digit ICD10 codes. This exercise is intended not merely to obtain better estimates of SARS-CoV-2 attributable deaths in India, but to explore the value of ex post verbal autopsies as a method of measuring the burden of the pandemic in lower-income countries.

Methods

Sample and data. As a first step, we create two samples: a sample of the Indian population alive at some point from 2018-2021 and a sample of Indians who died between 2018-2021. The first sample serves as a denominator and the second sample as a numerator in the estimation of mortality rates. The latter also serves as a sample frame for verbal autopsies.

Both living and deceased samples are drawn, in part, from a panel survey called the Consumer Pyramids Households Survey (CPHS), produced by the Centre for Monitoring Indian Economy (CMIE), Mumbai, India. This is a panel data set on 236,908 unique households intended to be representative of urban and rural areas of all states except those in the northeast of

India (Vyas 2020). The CPHS attempts to survey each household every 4 months (a round), with a representative quarter of sample households surveyed each month. The survey spans the period 2014-2022. While the CPHS is primarily used to measure the income and consumption of household members, it also updates household rosters in each round, allowing one to track those who are living and deceased. Supplement Section I provides more details on the CPHS survey.

Our sample of living Indians is the set of all living persons surveyed by CPHS between 2018-2021, a period that covers 2 years before the pandemic and 2 years during it.

Ideally, our sample of deceased persons would include all persons who would otherwise be in the living sample during the same subset of years (2018-2021), but who died in that period. However, our actual sample of deceased persons is a subset of this ideal sample. It is a subset because the CPHS data have three problems. First, these data do not report the cause of death. Second, they are over-inclusive: some deaths reported in 2018-2021 actually occurred before 2018. The CPHS surveys households every 4 months and only asks if a household member died, not when they died. Therefore, a household surveyed in January - April 2018 might report a death that occurred in September - December 2017. Third, they are under-inclusive: some deaths that occurred in 2018-2021 are not reported in CPHS until after 2021. For example, if a household is surveyed by CPHS in September 2021 but experienced a death in October - December 2021, that death would not be reported to CPHS until January 2022 and therefore omitted from our sample. The latter two problems are compounded if a household skips a CPHS round.

To address the cause-of-death and over-inclusive deaths problems, CMIE conducted WHO-sanctioned verbal autopsy (VA) surveys on all deaths reported on CPHS rosters from 2018-2021. These VA surveys asked the next-of-kin in each deceased CPHS sample member's

household the date of and circumstances of the decedent's passing. Cause-of-death was determined by a panel of doctors based on answers to a range of questions concerning the circumstances of the decedent's death. We address the over-inclusive problem by dropping all deaths that occurred before 2018.

The under-inclusive problems are addressed with a statistical “under-inclusivity adjustment” explained below, in the section on “Weights for deceased persons.”

Verbal autopsy survey and cause of death. CMIE conducted in-person visits to each household that reported a death in the CPHS survey during 2018-2021. At each household, relatives of the deceased person were asked to complete an electronic verbal autopsy (e-VA) survey. The e-VA employed the 2016 WHO Verbal Autopsy Standard tool, version 1.5.1.,¹² for neonates, children, adults, and stillbirths. The main change is that a few questions are added to identify symptoms associated with SARS-CoV-2. Our e-VA surveys are available as supplementary files. The e-VA software was run on laptops. Data were validated as they were entered; random re-sampling was employed for quality control. Details on the instrument are provided in a prior publication.¹³

Each death was assigned an International Classification of Diseases (tenth edition, ICD10) code that we treat as the cause of death by the Centre for Global Health Research, under the oversight of Prabhat Jha and Leslie Newcombe. The WHO added 3 codes to the ICD-10 classification to capture SARS-Cov-2 in 2021: U07.1, U07.2 and U08 (<https://icd.who.int/browse10/2019/en#/>). This assignment was done by two physicians on the basis of their review of the e-VA answers for the decedent. The two physicians we selected from a panel of physicians specially trained on death certification and ICD coding of verbal autopsies. Following the same procedure as the Indian Million Death Survey,^{14,15} these physicians coded

deaths independently and anonymously, and a third physician adjudicated any differences between the coding of the first two physicians, whose identities were masked.

We aggregate causes of deaths (aside from SARS-CoV-2) using a standard death classification from the Indian National Burden Estimates¹⁴ and aligned with the WHO Global Health Estimates.¹⁶ We classify deaths in our sample as caused by SARS-CoV-2 in 2 steps. First, if the 2-digit ICD10 code assigned to a decedent from the VA survey is U07 or U08, we call it a “potential SARS-CoV-2” death. This assignment is over-inclusive because U08 includes non-SARS-CoV-2 deaths (e.g., vaping-related disorders). Moreover, there may be errors in the verbal autopsies coding of cause of death, causing false positives (non-SARS-CoV-2 deaths coded as SARS-CoV-2) and false negatives (SARS-CoV-2 deaths reported as non-SARS-CoV-2). Second, to address either cause of over-inclusion, we calculate “SARS-CoV-2-attributable” excess deaths by subtracting the average annual rates of potential SARS-CoV-2 deaths in 2018-2019 from the average annual rates of those deaths in 2020-2021.

Outcomes. Our primary outcomes include the number of deaths and death rate attributable to all causes and to SARS-CoV-2, the top 5 other cause of death, and to other causes, both in our completed VA survey sample and in the population. Our primary outcomes also include the excess number of deaths attributable to all causes and to these same specific causes during the pandemic in the population. Our secondary outcomes are the these the primary outcomes, but for specific subpopulations: 5 different age groups, males and females, rural and urban residents, and households in each of 4 income quartiles.

Dates and date of death. We count time in months. We define the pandemic to be the period from January 2020 to December 2021. India’s national lockdown, which ran from March 24 to May 31, 2020, is coded as occurring during April and May 2020. Graphs mark September

2020 and April 2021 as the peak of India's first and second COVID waves based on peaks in officially reported infections ⁴.

The date of death is obtained in two steps. The CPHS provides the date that a household was surveyed and *reported* a member (the decedent) as having died since the last CPHS survey. This CPHS report date is later than the date of death because CPHS surveys households every 4 months and asks *if* a household member died, not when they died. Therefore, our first step is that CMIE, while conducting the verbal autopsy or a follow-up phone survey, asked each decedent's family the specific date of the decedent's death. In 1,569 cases, the household incorrectly remembered the decedent's date of death and gave a date *after* the CPHS report date, which is not possible. Our second step is that, in these 1,569 cases, we list the month of death as the month that is the midpoint between the (a) first month that the decedent's household answered a CPHS survey and reported the decedent as having passed and (b) the last month that households answered a CPHS survey and did not report that the decedent had passed. We call this date the midpoint date of death.

In the Supplement, we report an alternate set of results that equates the date of death for each decedent with the midpoint date of death, ignoring the specific dates households gave during the VA survey.

Statistical methods. The death rate from SARS-CoV-2 in a period is defined as the number of people who died from the disease (numerator) divided by the number of living people in that period plus the number of people who died in that period (denominator). The death rate may be unweighted, in which case it provides estimates about the sample. If it is weighted appropriately, it can provide estimates about the population. Using this framework, we provide information on the inputs into estimates of the death rate.

Weights for living persons. The sample of living persons, on its own, is not representative of either states or the nation. CPHS randomly samples a fixed number of households within strata defined by state, urban status, and settlement size; moreover, CPHS is unable to sample in parts of northeast India that suffer armed conflict. The resulting sample is not proportional to population at the state or district level, and thus at the national level given population variation across subnational units. Moreover, the living sample has survey non-response rates that vary across geography.

To extrapolate from our representative sample to the whole population of India, we need sampling weights. For each living person in our sample, CPHS provides weights are a composite of inverse probability sampling weights and adjustments for non-response in specific survey rounds ¹⁷. Details on these weights for the living population are provided in the Supplement. These “CPHS weights” permit the estimation of summary statistics that are unbiased for the living Indian population at the national, rural, and urban levels for each month. We also sum these “CPHS weights”, which determine the number of people each living respondent represents, across all living CPHS respondents in each month to estimate the total living population in each month.

Weights for deceased persons. The sample of deceased persons suffers the same two problems with representativeness as the living population. Weights are required to generate estimates of the deceased population for states or the nation.

There are three hurdles to the generation of weights for decedents in our sample. First, CPHS does not provide composite weights for deceased sample members for the month in which they are reported to CPHS as having died because they do not have an external estimate of local deaths. To address this problem, we impute “CPHS weights” for decedents in one of two ways.

One way is to assign the decedent a weight equal to the average CPHS weight of living members in the decedent's household in the month of death. If the decedent has no other household members surveyed that month, a second way is to assign them the average CPHS weight of other living respondents in CPHS that month. We use these imputed deceased weights to calculate population-level means for and totals of deceased persons.

The second hurdle is that, because households may be surveyed by the CPHS months after a member has died, our sample of deaths reported to the CPHS may miss deaths that occurred before 2022 but had not been reported to the CPHS by the end of 2021. We address this problem in two steps. Step one is to use the first 16 months of the VA survey sample to estimate the probability $p(t)$ that a death in a given month is reported to CPHS within t months, where t ranges from 1 to 32. Step two is, for each death that is reported in t months before December 2022, we create a scaling-up factor, equal to $1/p(t)$. This scaling-up factor, which we call an “under-inclusivity adjustment”, can be interpreted as the number of actual CPHS sample deaths that each reported CPHS death represents. Additional details on this adjustment are provided in the Supplement.

The third hurdle is that not all CPHS sample households with a reported decedent respond to our VA survey. Moreover, those who do respond may not be representative of those who do respond to the VA. To address this non-random undercounting of the numerator due to non-response to the VA, we compute an adjustment in three steps. First, we estimate a prediction model for whether a household with a decedent reported to the CPHS responds to the VA survey. This prediction model includes household demographic and economic covariates, as well as strata fixed effects and the CPHS weights. Second, we create a propensity-score weight for each decedent whose household responded to the VA survey. This “VA non-response”

weight is equal to the reciprocal of the predicted probability that the decedent's household responds to the verbal autopsy. Additional details on this weight are provided in the Supplement.

Our composite weight for all decedents who respond to the VA survey is the product of the CPHS weight, the under-inclusivity adjustment, and the VA non-response weight.

Count of living persons. The denominator includes a count of living persons in the CPHS sample. The unweighted count of all living persons in a month is the set of all living persons whose households responded to a CPHS survey in that month. The response rate for the CPHS survey in a month is the fraction of households that CPHS attempted to survey in a month that responded to that survey in that month. Our estimate of the number of all living persons in the population in a month is the sum of the weights of all persons who responded to the CPHS in that month. These weights sum up to the number of individuals the CPHS projects to be in the population in a given month.

Count of deaths. The unweighted count of all-cause or cause-specific VA deaths in a month is estimated with the total number of decedents in the actual VA sample who died from the designated cause(s) and whose date of death is that month. We estimate the all-cause or cause-specific number of deaths in the CPHS sample by calculating the weighted sum of deaths reported in the VA sample from the designated cause(s) in that month, where the weight on each death is the product of the associated VA non-response weight and the under-inclusivity adjustment. We estimate the number of all-cause or cause-specific deaths in the population with a similar weighted sum, except that each weight is the product of the VA non-response weight, the under-inclusivity adjustment, and the imputed CPHS weight for a decedent. █

Count of excess deaths. The number of all-cause or cause-specific excess deaths in the CPHS sample in a month during the pandemic is the all-cause or cause-specific number of deaths in the CPHS sample in that month minus the average number of such deaths each month during 2018-2019.

Death rate. The unadjusted all-cause death rate in the CPHS sample in a period is estimated with the fraction of the CPHS sample that dies during that period. If the period is, for example, a month, the numerator is the number of deaths in the VA sample in a month and the denominator is the number of CPHS living respondents in a month. Note that CPHS sample members who die in that month are not included in the denominator, producing a slight overestimate of the all-cause death rate in the sample. We discuss this problem further when explaining our estimates of population-level death rates below.

The VA non-response adjusted death rate modifies the numerator to be the sum of non-response adjustments for each decedent in a month. The under-inclusivity adjusted death rate modifies the numerator further: it is the product of each decedent's VA non-response weight and under-inclusivity adjustment, summed over all VA decedents in a month.

Our estimate of the death rate in the population modifies both numerator and denominator. The numerator is the product of the VA non-response adjustment, the under-inclusivity adjustment, and the imputed composite CPHS weight for each decedent, summed over decedents who die in a month. The denominator is the sum of composite CPHS weights in the CPHS sample of living persons that month.

We do not add the numerator (which captures deaths in the population) to this sum because CPHS's estimate of the population, and thus its inverse probability weights, are based on a projection about India's population that does not include an adjustment for the pandemic. That

is, the weights assume pre-pandemic trends in mortality rates and no additional deaths during the pandemic. This means that our estimate of population-death rates produces an overestimate of death rates: the numerator includes predicted mortality from pre-pandemic trends plus the excess deaths during the pandemic, while the denominator includes living persons and excess deaths during the pandemic but not predicted mortality from pre-pandemic trends. However, adding all pandemic-period deaths to the denominator would lead to an underestimate of the death rate. While it would offset the omission of predicted pre-pandemic mortality in the denominator, it would add pandemic excess deaths a second time, leading to a double-counting of those excess deaths. Although both biases are likely small because death rates are low, we judge that an overestimate is more conservative.

The analogous cause-specific death rates use only VA sample members with the designated cause of death in the numerator.

Share of deaths attributable to a cause. The share of deaths that are due to a given cause in a month is estimated as the ratio of (a) the cause-specific fully-adjusted death rate in a month and (b) the all-cause fully-adjusted death rate in a month.

Excess-death rate. The population excess death rate attributable to a cause is defined as the increase in the population death rate from that cause during the COVID pandemic. The numerator of this rate is estimated by subtracting (a) the average fully adjusted death rate from that cause in 2018-2019 from (b) the fully adjusted death rate from that cause during the pandemic. The denominator of that rate is estimated by subtracting (a) the average fully-adjusted all-cause population death rate in 2018-2019 from (b) the fully-adjusted all-cause death rate during the pandemic.

Heterogeneity in death rates. We estimate variation in SARS-CoV-2 and non-SARS-CoV-2 death rates by attributes such as individual demographics, household geography, and household income via linear regression analysis. We employ a sample of all living CPHS respondents and VA decedents in the period 2020-2021. Data are monthly. A person is recorded as having died in a month by an indicator variable.

We regress an indicator for whether the sample member died in a month from a cause, i.e., was in the VA sample and died in that month due to the cause, against indicators for the decedent's demographics, locations, and/or household income. Decedent demographics are captured by indicators for 20-year wide age bins and for their sex. Location is captured with an indicator if the household resides in a rural village (using the 2011 Indian Census' definitions for village). Household income is measured by the local income quartile of the decedent's family. That indicator is generated using the total income of the household of the decedent in all of 2018-2019 to determine quartiles of household income for local areas defined by state and urban status. We include weights that make the regression population-representative. We report regression results that include one set of factors at a time, not all factors at once, so that no category, e.g., females or rural, is omitted for each set. However, we examine a regression with all sets of factors included at once to check if, e.g., the correlation of income on death rates is not driven by age. We multiply coefficients by 12 to ensure that they represent annual rather than monthly death rates.

We determine whether demographic, geographic and income factor have a different correlation with SARS-CoV-2 versus non-SARS-CoV-2 death rates by estimating a regression where the dependent variable is whether the sample member died (regardless of cause) on factors, an indicator for whether the cause of death was SARS-CoV-2, and the interaction of

factors with an indicator for whether SARS-CoV-2 was the cause of death. Furthermore, we include weights that make the regression population-representative and multiply coefficients by 12 to ensure that they represent annual rather than monthly death rates.

We show further geographical correlates of SARS-CoV-2 death rates by calculating annual SARS-CoV-2 death rates for each Indian state for the period 2020-2021 and plotting these using a choropleth map of India.

Results

Sample size and response rates. *Living sample.* The sample of (living) persons who ever respond to the CPHS contains 892,905 unique persons across 174,003 unique households in the period 2018 to 2021. Because individuals and households rotate into and out of the sample, this amounts to 645,050 unique persons across 131,200 households on average per round in 2018-2021.

Figure 1 plots the number of living individuals (blue line) who respond to a CPHS survey each month from 2018-2021. The household-level response rate to the CPHS survey averaged 85.0% in 2018-2019. That rate fell in March 2020, when India declared a lockdown. It reached a perigee in April 2020 at 30.9% before the peak of India's first COVID Wave. It substantially recovered between India's first 2 waves, but briefly dropped again, to a low of 46.1% in May 2021, just after the peak of the second wave. The green line presents the sum of composite CPHS weights, which account for sampling probabilities and CPHS non-response rates, and thus the total population represented by the CPHS sample. This population is linearly increasing and in the range of 1.4 billion persons during the sample period.

Deceased sample. The CPHS decedent sample contains 30,447 unique individuals that are reported between 2018-2021 as having died. These individuals appear in the CPHS until the date that their death is reported by living household members in the CPHS survey.

Although VA surveys were attempted for each of these decedents, only 22,178 surveys were completed and included in the actual VA sample, a response rate of 72.8%. Non-response to the VA is not random. Households associated with the CPHS sample of deceased persons that respond to the VA are more likely to, among other things, live in a city ($p < 0.001$), have fewer household members ($p < 0.001$), have more children ($p < 0.001$), have fewer adult males ($p < 0.001$), not be from a backward caste ($p < 0.001$), self-report poorer health ($p < 0.001$), and not be Hindu ($p = 0.002$) or Muslim ($p < 0.001$) ([Supplement, Table S1, column 1](#)). Using propensity score weights (based on the model in column 1, but replacing the rural indicator with strata fixed effect) eliminates these imbalances between VA-responsive and non-responsive households ([Supplement, Table S1, column 3](#)).

The CPHS decedent sample is over-inclusive, i.e., includes some deaths that occurred prior to 2018 but were reported after 2018. Among decedents for which a VA was completed, 1,719 (7.8%) had dates of death prior to 2018. Our actual VA sample, which excludes these persons, is 20,453.

The CPHS is also under-inclusive because deaths are reported to the CPHS some months after they occur, a date measured in the VA survey ([Supplement, Figure S1](#)). Our adjustment for under-inclusivity accounts for 32-month window of reporting following the date of death. Of the deaths that are reported in this window, 52.7% are reported to CPHS within 4 months of occurring, a lag equal to the period between CPHS surveys of each household; the rate of reporting levels off so that it takes nearly 28 months to reach 98.3% reporting in the 31st month.

Figure 2 plots the number of deaths (orange line) in the actual VA sample that occurred each month. Our use of propensity score weights to correct for VA non-response (green line) increases our estimate of the total number of deaths reported in the CPHS by 39% in the median month. Our adjustment for the delay in reporting deaths to the CPHS survey (i.e., under-inclusivity) increases our estimate of the number of deaths – not just reported deaths – in the CPHS sample by 17% in the median month of the sample period, but by 10.6 times on average in the last 4 months of that period. That spike may be an early indicator of India’s third wave (the Omicron variant), which peaked in January 2022. The purple line reports the sum of weights for each death, thus the estimated total number of deaths in the population (right y-axis) from which CPHS samples, accounting for sampling probabilities and CPHS non-response rates.

Cause-specific death count. **Figure 3A** plots our estimate of the number of deaths in the CPHS living sample over time that are attributable to SARS-CoV-2, the 5 leading causes of death prior to the pandemic (cardiovascular diseases, accidents, respiratory disease, diarrheal disease, and malaria), and deaths from other causes. These counts adjust for non-response to the VA survey and delayed or under-inclusive reporting in the CPHS, but not weights that would make them representative of deaths in the population. The count of SARS-CoV-2 deaths is non-zero in 2018-2019. However, we code both U07 and U08 ICD10 codes as SARS-CoV-2-related, even though U07 deaths include vaping-related disorders, which predate the pandemic.

During the pandemic, deaths coded as U07 or U08 (thus likely SARS-CoV-2) in the CPHS sample spike to (a) to 283 during the lockdown (gray region) and peak just after it, (b) to 231 during India’s first wave (orange vertical line), and (c) to 380 just after India’s second wave (purple vertical line). Cardiovascular and non-specific “other” deaths also spike at the start of 2020 and during the lockdown, around the two COVID waves, and especially at the end of 2021.

Figure 3B reports excess deaths by cause defined as deaths by cause from panel A minus the average number of deaths from that cause in 2018-2019. This highlights the causes of deaths that rise the most during the pandemic. SARS-CoV-2 excess deaths spike more than excess deaths from any other cause during and just after the lockdown (240 deaths), during wave 1 (188 deaths), and just after wave 2 (338 deaths). Respiratory excess deaths fall by 57.4 per month ($p < 0.001$) on average during the pandemic. Due to our under-inclusivity adjustment, “other” cause excess deaths spike at the end of 2021, perhaps related to India’s third COVID wave in early 2020.

All-cause and cause-specific death rate. **Figure 4** plots the monthly all-cause death rate. The orange line gives the raw death rate. The green line adjusts for VA non-response and CPHS under-inclusivity. The red line adds CPHS sampling weights to estimate the death rate in the population. Adjustments have little effect prior to pandemic. The green and red line are similar, suggesting that CPHS sampling probabilities matter less to pandemic death rates than VA non-response rates and CPHS under-inclusivity. Because under-inclusivity weights are large mainly in the last 4 months of 2021, the VA non-response weights have the largest influence on pandemic period death rates. Looking at the red line, the estimated population death rate spikes in January 2020, during the lockdown in May 2020, May 2021 and December 2021 to annualized rates of 3.0, 4.6, 3.0, and 2.1%, respectively.

Figure 5A plots cause-specific estimated population death rates for SARS-CoV-2, 5 leading causes of death prior to the pandemic, and other causes of death. During the pandemic, the SARS-CoV-2 death rate peaks at annualized rates of 0.8% in May 2020 during lockdown, 0.9% in June 2020 just before the peak of wave 1, and 1.1% in May 2021 just after the peak of wave 2. These peaks are not proportional to the estimated number of deaths in the CPHS sample

because sample members who died at different times may have had different sampling weights. Moreover, the weight of the denominator for the death rate rises as the population of India grows over time. Cardiovascular death rates spike to 1.1% in January 2020, 1.2% in May 2020, and 0.7% in November 2021. Death rates for “other” causes spike to 1.2% in May 2020 and 1.4% in December 2021. Although respiratory deaths decline during the pandemic, respiratory death rates are not significantly lower during the pandemic.

Figure 5B plots cause-specific death rates but as an index value, created by dividing cause specific death rates by average death rate for that cause in 2018-2019. One purpose of this index is to net out the effect of vaping-related disorders in the U07 and U08 ICD10 codes so that those codes place greater weight on SARS-CoV-2.

These index values highlight that the spike in death rates was more prominent for SARS-CoV-2 than for cardiovascular or other causes just after the lockdown and especially after wave 2. Moreover, the figure shows that cardiovascular and other cause death rates do not spike after the first few months of the pandemic. After the lockdown they tend to decline. An exception is the last to months of 2021, but the correction for under-inclusivity is driving the spikes. Without that very large adjustment, the spike at the end of 2021 vanishes.

Share of deaths due to SARS-CoV-2. SARS-CoV-2 becomes a large portion of all deaths starting in 2020 (**Figure 6**). Its relative importance grows steadily until it spikes in June 2020 and April 2021, when it accounts for over 23.3% and for nearly 35.8% of deaths. After May 2021, the share of deaths attributable to SARS-CoV-2 diminishes.

The all-cause annual death rate in 2020-2021 is 81.19% higher than the annual death rate in 2018-2019 (**Table 1**). SARS-CoV-2 is associated with 17.93% of all deaths during pandemic, and 33.15% of excess deaths during the pandemic. Cardiovascular disease is associated with

27.18% of all deaths and 23.15% of the increase in all-cause death rates during the pandemic. Respiratory disease-related deaths fall by an amount equal to 1.52% of excess deaths during the pandemic. Finally, accidents and tuberculosis are responsible for 8.74 and 5.32% of excess deaths during the pandemic.

SARS-CoV-2 death rate by demographic, geographic and economic group. The death rate from SARS-CoV-2 is significantly higher in higher age groups. It is highest among individuals who were above the age of 80 when they died, followed by those between ages 61-80 (Figure 7a, blue dots). However, because so few people are over 80, the vast bulk of SARS-CoV-2 deaths were among those between ages 61-80 (blue bars). Death rates from non-COVID causes are also significantly higher in higher age groups (Figure 7b, blue dots). Indeed, the age gradient on non-COVID death rates is significantly greater than the age gradient on SARS-CoV-2 death rates ($p < 0.001$).

The SARS-CoV-2 death rate is higher among females, among urban residents, and in the highest income quartile, but not significantly so (Figure 7a). While urban residents have higher death rates, total deaths are greater in rural areas because there are more residents in rural areas. Non-COVID death rates follow different patterns than SARS-CoV-2 (Figure 7b). They are higher among males and rural residents, though not significantly so. The differences between non-COVID and SARS-CoV-2 death rates are significantly different for rural residents, but not for males or different income quartiles.

Figure 8 presents the annual SARS-CoV-2 excess death rates for different states during the pandemic. Death rates tend to be lower in the Gangetic plain (e.g., Uttar Pradesh 0.13%; Bihar 0.07%; Jharkhand, 0.07%) and highest in Goa (2.13%), Meghalaya (1.8%), and Tripura (1.42%).

Discussion

We completed verbal autopsy (VA) surveys on a sample of 20,453 persons who are reported in 2018-2020 as having died to determine the cause of death. VA surveys were attempted on a representative sample of 30,447 deaths in the CPHS sample during this period. The final VA sample reflects a 72.8% response rate on attempted surveys, plus the elimination of 7.8% of completed surveys because the deaths actually occurred prior to 2018.

The timing of spikes in COVID waves differs somewhat from official reports. Official reports of SARS-CoV-2-attributed deaths spike during India's first (September 2020) and second (April 2021) waves. VA surveys confirm the spike during wave 2. However, the 2020 spike occurs at the tail end of the lockdown (May-June 2020) rather than during wave 1. Our findings are consistent with the finding that SARS-CoV-2 rates spiked during the lockdown, especially among the urban poor¹⁸.

The discrepancy between official reports and our data have two possible explanations that are not mutually exclusive. Official reports are based on SARS-CoV-2 tests by hospitals on individuals prior to death, and are reported by hospitals to the government. Errors are possible because official statistics only include deaths in hospitals, not the community, because of flaws in the reporting process, or because governments may decide not to report deaths¹⁹. Our estimated dates of death might be the result of recall error among surviving family members, the respondents to verbal autopsy surveys. However, this recall error may be positive or negative and is likely independent across households; so, they may converge to zero in larger samples.

Death rates from SARS-CoV-2 disproportionately spike only during wave 2. SARS-CoV-2 death rates spike at the end of the lockdown and during wave 2. However, death rates

from other causes – notably cardiovascular disease and all “other” causes also spike during the lockdown. This gap between SARS-CoV-2 and non-COVID death rates may explain why wave 2 seemed especially stark in India.

Death rates do not translate directly into deaths. SARS-CoV-2 death rates increase significantly with age, with populations above age 80 having the highest death rates. However, there were fewer deaths of those above 80 because there are far fewer people above 80 than between 60 and 80 in the population. Likewise, while SARS-CoV-2 death rates were greater in urban areas than in non-urban areas, there were more SARS-CoV-2 deaths in rural areas because the rural population is larger than the urban population.

SARS-CoV-2 deaths are surprisingly less correlated with age than non-COVID deaths. SARS-CoV-2 death rates increase significantly with age, with populations above age 80 having the highest death rates. However, non-COVID death rates also increase with age. Indeed, the difference between the over-80 death rate and the under 20 death rate is significantly greater for non-COVID causes of death than for SARS-CoV-2 ($p < 0.0001$).

Income does not have a clear impact on SARS-CoV-2 deaths. While ⁶ finds that all-cause excess death rates are significantly, positively correlated with income during the pandemic, we find that the correlation between the SARS-CoV-2 death rate and income is not significant.

SARS-CoV-2 explains 17.93% of deaths and 33.15% of excess deaths during the pandemic. Cardiovascular deaths are responsible for 23.15% of excess deaths during the pandemic. However, some of these individuals may have died in part due to SARS-CoV-2. Individuals with cardiovascular disease are at greater risk of mortality conditional on SARS-CoV-2 infection ²⁰. Moreover, SARS-CoV-2 may cause cardiac damage ^{21,22}. So, it is possible SARS-CoV-2 deaths are underestimated by the VA survey.

Respiratory deaths fell during the pandemic, by an amount equal to 1.52% of excess deaths. It is possible that some of the increase in SARS-CoV-2 deaths are actually mis-coded respiratory deaths. That suggests that we may overestimate SARS-CoV-2 excess deaths, though the magnitude of the overestimate is small.

This paper has several limitations. The VA survey suggests there were death attributable to ICD10 categories U07 and U08 in 2018-2019. While we call U07 and U08 deaths as SARS-CoV-2 deaths, U07 includes deaths associated with vaping related disorders. This may be why Figure 3A reports a positive number of SARS-CoV-2-labeled deaths in 2018-2019. Vaping was a public health crisis prior to the pandemic²³, including in India²⁴. We attempt to correct for this by converting cause-specific death rates into index values in Figure 5B and by calculating excess death rates in Table 1: both these steps net out average death rates in 2018-2019. However, vaping may remain partly responsible for the rise in excess SARS-CoV-2-related deaths in 2020-2021 because vaping may compound the risk from SARS-CoV-2 infection²³. In general, one must be cautious about relying on mono-causal explanations of death.

A related issue is that SARS-CoV-deaths might be miscategorized as other respiratory disease or infectious disease deaths. To some extent this type of error is limited by the fact that surveyors and coders were aware of revisions to the VA survey and to the CPT codes that account for the new SARS-CoV-2 diagnosis and the need to be particular in capturing those deaths. Including deaths potentially misclassified as respiratory, ill-defined, unknown fever, or other infectious diseases in the count of SARS-CoV-2 deaths increases the fraction of all deaths in 2020-2021 that are potentially SARS-CoV-2 deaths from 17.93% to 32.68% (Table 1).

However, this re-classification only bumps up the fraction of excess deaths that are attributable

to SARS-CoV from 33.15% to 35.40% because the additional classifications play a small role in the spike in deaths in 2020-2021.

Second, the completed VA sample requires statistical corrections to be considered representative, but those corrections may be imperfect. The representative sample of CPHS deaths that is used as the sample frame for the VA is missing deaths which occur during between 2018-2021 but are not yet reported to CPHS. We correct that with a factor that accounts for the statistical delay in reporting deaths. But that correction can only correct for deaths that are ultimately reported in 48 months; we have no information and cannot correct for deaths that are reported more than 4 years after occurring. Moreover, our correction leads to surprisingly sharp increases in death rates from “other” causes, which are hard to understand.

In addition, the households that respond to the VA survey are not a random sample of all CPHS households that reported a death in 2018-2021. We correct for this with a propensity score weight, which does a reasonable job eliminating imbalance in attributed between households that do and do not respond to VA surveys. However, our adjustment cannot correct for imbalance on variables that are not used in our propensity score estimation model.

The sample of living persons that informs the denominator of our estimate of death rates may also suffer selection bias. While we address these with weights that address population sizes and non-response, the latter correction assumes random non-response. To the extent that non-response rate is not at random, overall or at the cause-specific level, our estimates of death rates will be biased.

Third, the verbal autopsy survey we employed was well-validated and used prior to the pandemic^{12,13}. However, the procedure for mapping answers to diagnostic codes had to be modified for a new diagnosis, SARS-CoV-2. The specific modification we employed has not yet

been validated – although other modifications to the WHO VA instrument to address COVID have been ^{25,26}. Nevertheless, assessments of VA modifications to capture SARS-CoV-2 show there is error and diagnosis should be complemented with minimally invasive techniques such as ultrasound when the decedent's cadaver is available ²⁷.

ARTICLE IN PRESS

References

- 1 De-Giorgio, F. *et al.* Dying “from” or “with” COVID-19 during the Pandemic: Medico-Legal Issues According to a Population Perspective. *International Journal of Environmental Research and Public Health* **18** (2021).
- 2 Ioannidis, J. P. A. Over- and under-estimation of COVID-19 deaths. *European Journal of Epidemiology* **36**, 581-588 (2021). <https://doi.org/10.1007/s10654-021-00787-9>
- 3 Levin, A. T. *et al.* Assessing the burden of COVID-19 in developing countries: systematic review, meta-analysis and public policy implications. *BMJ Global Health* **7**, e008477 (2022). <https://doi.org/10.1136/bmjgh-2022-008477>
- 4 Johns Hopkins University & Medicine. COVID-19 Dashboard by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University (JHU). (2021).
- 5 Jha, P. *et al.* COVID mortality in India: National survey data and health facility deaths. *Science* **375**, 667-671 (2022). <https://doi.org/10.1126/science.abm5154>
- 6 Malani, A. & Ramachandran, S. Using household rosters from survey data to estimate all-cause excess death rates during the COVID pandemic in India. *Journal of Development Economics* **159**, 102988 (2022). <https://doi.org/https://doi.org/10.1016/j.jdeveco.2022.102988>
- 7 Banaji, M. & Gupta, A. Estimates of pandemic excess mortality in India based on civil registration data. *PLOS Global Public Health* **2**, e0000803 (2022). <https://doi.org/10.1371/journal.pgph.0000803>
- 8 Anand, A., Sandefur, J. & Subramanian, A. Three new estimates of India’s all-cause excess mortality during the COVID-19 pandemic. (2021).
- 9 Leffler, C. T., Lykins, V. J., Das, S., Yang, E. & Konda, S. Preliminary Analysis of Excess Mortality in India During the COVID-19 Pandemic. *The American journal of tropical medicine and hygiene* **106**, 1507-1510 (2022). <https://doi.org/10.4269/ajtmh.21-0864>
- 10 Rao, C. & Gupta, M. The civil registration system is a potentially viable data source for reliable subnational mortality measurement in India. *BMJ global health* **5**, e002586 (2020). <https://doi.org/10.1136/bmjgh-2020-002586>
- 11 Gamio, L. & Glanz, J. in *New York Times* (2021).
- 12 WHO. *Verbal autopsy standards: the 2016 WHO verbal autopsy instrument*, <<https://www.who.int/standards/classifications/other-classifications/verbal-autopsy-standards-ascertaining-and-attributing-causes-of-death-tool>> (2016).
- 13 Carshon-Marsh, R. *et al.* Child, maternal, and adult mortality in Sierra Leone: nationally representative mortality survey 2018–20. *The Lancet Global Health* **10**, e114-e123 (2022). [https://doi.org/10.1016/S2214-109X\(21\)00459-9](https://doi.org/10.1016/S2214-109X(21)00459-9)
- 14 Menon, G. R. *et al.* National Burden Estimates of healthy life lost in India, 2017: an analysis using direct mortality data and indirect disability data. *The Lancet Global Health* **7**, e1675-e1684 (2019). [https://doi.org/https://doi.org/10.1016/S2214-109X\(19\)30451-6](https://doi.org/https://doi.org/10.1016/S2214-109X(19)30451-6)
- 15 Dhingra, N. *et al.* Adult and child malaria mortality in India: a nationally representative mortality survey. *The Lancet* **376**, 1768-1774 (2010). [https://doi.org/10.1016/S0140-6736\(10\)60831-8](https://doi.org/10.1016/S0140-6736(10)60831-8)

- 16 WHO Global Health Observatory. *Global health estimates 2019: deaths by cause, age, sex, by country and by region, 2000–2019*,
<<https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates/gh-leading-causes-of-death>> (2020).
- 17 Vyas, M. Weights. (Center for Monitoring Indian Economy, 2020).
- 18 Sheng, J., Malani, A., Goel, A. & Botla, P. Does mobility explain why slums were hit harder by COVID-19 in Mumbai, India? *Journal of Urban Economics* (2021).
- 19 Chatterjee, P. Is India missing COVID-19 deaths? *The Lancet* **396**, 657 (2020).
[https://doi.org/10.1016/S0140-6736\(20\)31857-2](https://doi.org/10.1016/S0140-6736(20)31857-2)
- 20 Shi, S. *et al.* Association of Cardiac Injury With Mortality in Hospitalized Patients With COVID-19 in Wuhan, China. *JAMA Cardiology* **5**, 802-810 (2020).
<https://doi.org/10.1001/jamacardio.2020.0950>
- 21 Niazi, S., Niazi, F., Doroodgar, F. & Safi, M. The Cardiac Effects of COVID-19: Review of articles. *Curr Probl Cardiol* **47**, 100981 (2022).
<https://doi.org/10.1016/j.cpcardiol.2021.100981>
- 22 Terzic, C. M. & Medina-Inojosa, B. J. Cardiovascular Complications of Coronavirus Disease-2019. *Phys Med Rehabil Clin N Am* **34**, 551-561 (2023).
<https://doi.org/10.1016/j.pmr.2023.03.003>
- 23 Shin, Y. M., Hunt, D. P. & Akwe, J. An Epidemic Supplanted by a Pandemic: Vaping-Related Illness and COVID-19. *South Med J* **115**, 8-12 (2022).
<https://doi.org/10.14423/smj.0000000000001344>
- 24 Pettigrew, S. *et al.* E-cigarettes: A continuing public health challenge in India despite comprehensive bans. *Prev Med Rep* **31**, 102108 (2023).
<https://doi.org/10.1016/j.pmedr.2022.102108>
- 25 Duarte-Neto, A. N. *et al.* Rapid Mortality Surveillance of COVID-19 Using Verbal Autopsy. *Int J Public Health* **66**, 1604249 (2021).
<https://doi.org/10.3389/ijph.2021.1604249>
- 26 Rosen, T. *et al.* Development of the Verbal Autopsy Instrument for COVID-19 (VAIC). *J Gen Intern Med* **36**, 3522-3529 (2021). <https://doi.org/10.1007/s11606-021-06842-1>
- 27 de Souza, P. M. M. *et al.* Validation of verbal autopsy and nasopharyngeal swab collection for the investigation of deaths at home during the COVID-19 pandemics in Brazil. *PLOS Neglected Tropical Diseases* **14**, e0008830 (2020).
<https://doi.org/10.1371/journal.pntd.0008830>

Declarations

Funding. The Centre for Monitoring the Indian Economy (CMIE), a commercial survey firm, conducts a household demographic and economic survey, the Consumer Pyramids Household Survey (CPHS) in India. De-identified data from the CPHS can be purchased for a subscription fee by any member of the public. The author is not affiliated with CMIE; he is an employee of the University of Chicago. The Becker-Friedman Institute and the Provost Global Faculty Award at the University of Chicago paid for the author's subscription to the de-identified CPHS data and research assistant time. This is the only funding the author obtained for this study.

Emergent Ventures and an anonymous donor commissioned CMIE to conduct a verbal autopsy survey in India based on the CPHS sample frame and make it available to the public. CMIE worked with Prabhat Jha and Leslie Newcombe at the Centre for Global Health Research (CGHR) to recruit and manage the physician panel that mapped map CMIE's verbal autopsy results to ICD10 codes. Emergent Ventures required that CMIE ensure that the de-identified CGHR mapping and identifiers that allows the data to be merged with de-identified CPHS data, without person identifiers, be released to the public without charge. The author obtained the de-identified verbal autopsy data from CMIE without charge.

Ethics approval and consent to participate. CMIE field teams obtained verbal informed consent from each adult next-of-kin respondent before conducting each verbal-autopsy survey. Participation was voluntary and could be declined at any time. CMIE, assisted by CGHR, complied with WHO norms for consent for verbal autopsies, and the WHO norms (see <https://www.who.int/standards/classifications/other-classifications/verbal-autopsy-standards-ascertaining-and-attributing-causes-of-death-tool>) comply with the Helsinki Declaration. CMIE

field teams likewise obtained informed consent before conducting the CPHS survey. CPHS consent also complies with the Helsinki Declaration. Each of these surveys is a commercial survey product available to the public without personal identifiers.

The IRB of Unity Health hospital system (Case no. REB #15-231) gave the Centre for Global Health Research (CGHR) approval (including Prabhat Jha and Leslie Newcombe) to create the physician panel that mapped map CMIE's verbal autopsy results to ICD10 codes that CMIE made available to the author for this study.

The University of Chicago IRB determined the author's study (Case No. IRB22-0119) was not human subjects research because the data were gathered by a commercial entity (not the author), the data are or will be publicly accessible, and no identifiable information is shared with the author. The author of this paper is an employee of the University of Chicago. He served as an unpaid advisor to CMIE for designing the sample frame for the verbal autopsy survey, but is not an employee or affiliate of CMIE. The author did not have access to any personally identifiable information from the survey.

Consent for publication. This study contains—and the author received—no personally identifiable data.

Availability of data and materials. De-identified data (including date of death, ICD10 codes, and basic socio-economic and demographic data) for each subject in the sample will be made publicly available on the author's Github page. These data may also be obtained directly from CMIE under the terms of the Emergent Ventures grant.

Competing interests. The author declares no competing interests.

Authors' contribution. The sampling frame for the verbal autopsy survey was determined by CMIE, with (unpaid) advice from AM. The verbal autopsy survey was designed

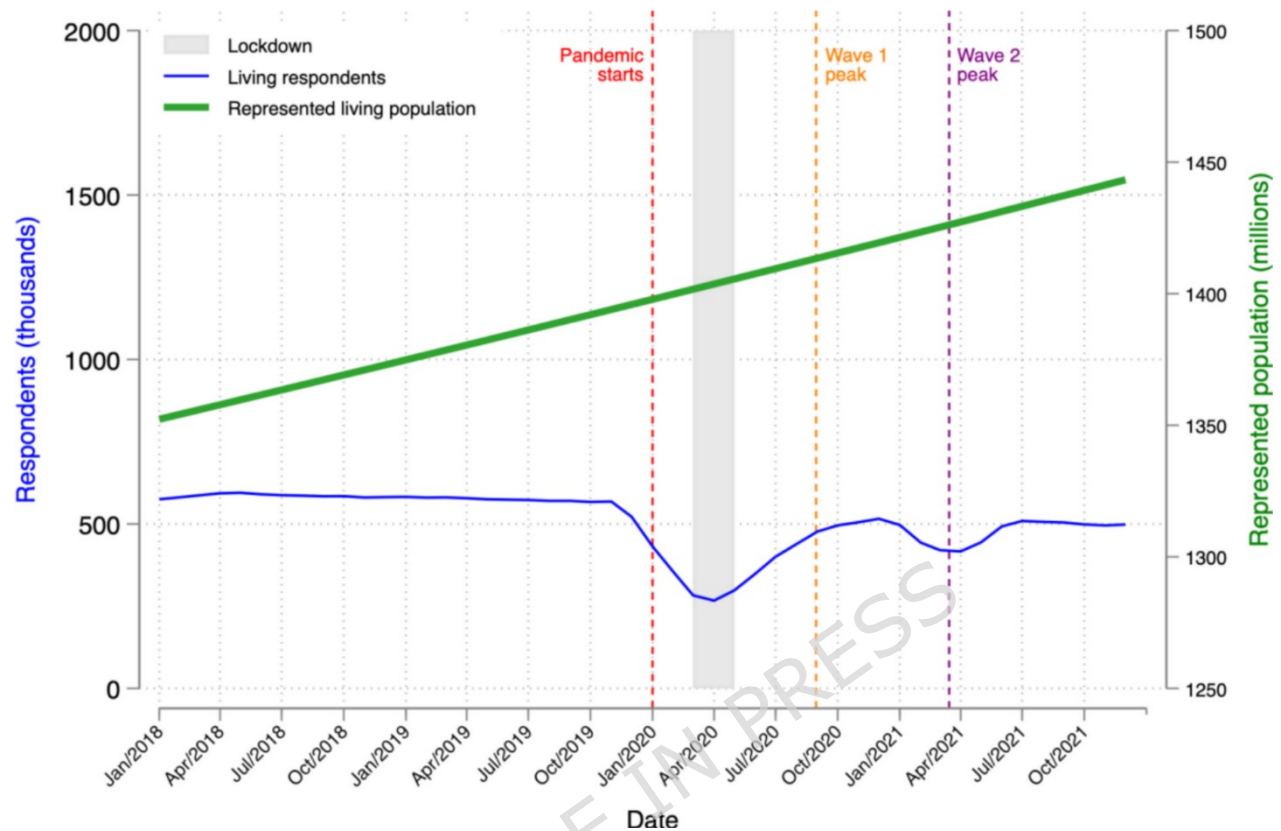
by the Centre for Global Health Research in Toronto, Canada. The surveys were administered by Centre for Monitoring Indian Economy in Mumbai, India. The surveys were processed and mapped to ICD10 codes by Centre for Global Health Research. The CPHS survey and its sampling frame was determined by CMIE. That survey was implemented by CPHS.

The design for the study in this paper was conceived by AM. The verbal autopsy data and the CPHS data on demographics and socioeconomics of households were merged by AM and a research assistant working for AM. Statistical analyses were designed by AM and implemented by AM and the research assistant on de-identified data. Writing was done by AM. The opinions expressed by the author of this manuscript do not necessarily reflect the opinions of the institution with which the author is affiliated.

Acknowledgements. The author the critical support of Mahesh Vyas at the Centre for Monitoring Indian Economy (CMIE), Prabhat Jha and Leslie Newcombe from the Centre for Global Health Research, and research assistance by Wanran Zhao. CMIE conducted the verbal autopsy surveys. Jha and Newcombe's team at CGHR mapped answers to the survey onto ICD10 codes. CGHR also helped provide us cleaned data on date of death and cause of death. We also thank Emergent Ventures and the anonymous donor that funded CMIE to help it create the CMIE Verbal Autopsy database for the public.

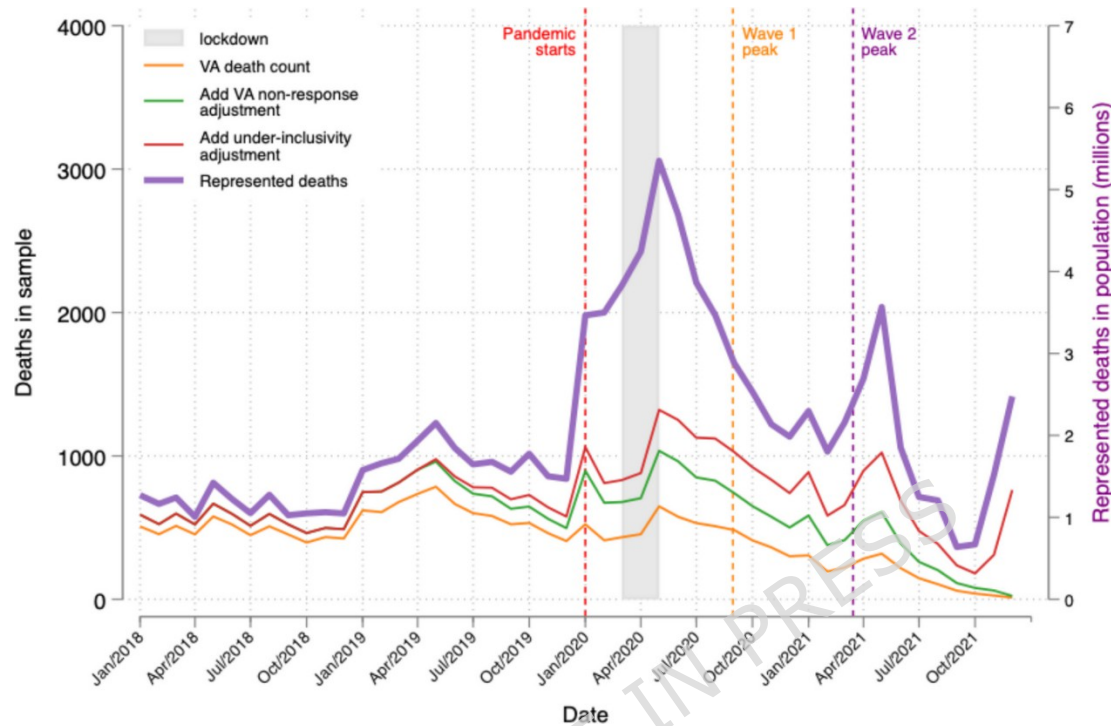
Figure legends

Figure 1. Living CPHS respondents and represented living population by month.



Notes. Blue line (left y-axis) plots the total number of individual respondents per month to CPHS survey. Green line (right y-axis) plots the sum of weights for these respondents. Weights account for both sampling probability and rate of non-response to the CPHS survey. So green line indicates the total population size represented by CPHS. Red dashed line marks the beginning of COVID pandemic; gray area marks the period of India's national lockdown; and the yellow and purple lines mark the peaks of India's first and second COVID waves.

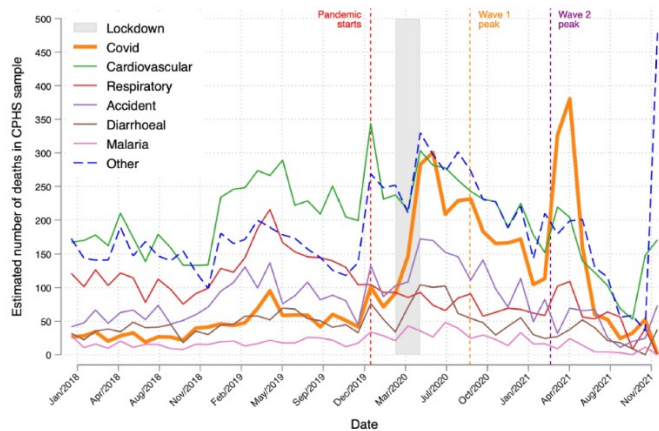
Figure 2. Counts of deaths by month, unadjusted in VA sample, adjusted for VA non-response, adjusted for under-inclusivity of CPHS, and adjusted to be population representative.



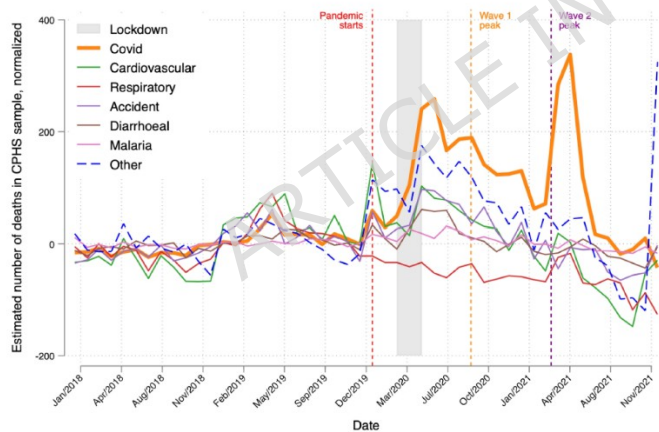
Notes. Orange line (left y-axis) plots the total number of individual decedents per month according to verbal autopsy (VA) survey. Green line (left y-axis) adjusts orange line to address non-response to the VA. Red line (left y-axis) adjusts green line to address delay in reporting deaths to CPHS (under-inclusivity). Purple line (right y-axis) plots the sum of weights for these decedents captured by red line.

Figure 3. Monthly death counts for SARS-CoV-2, 5 other leading causes of death, and other deaths, 2018-2021.

A: Absolute numbers

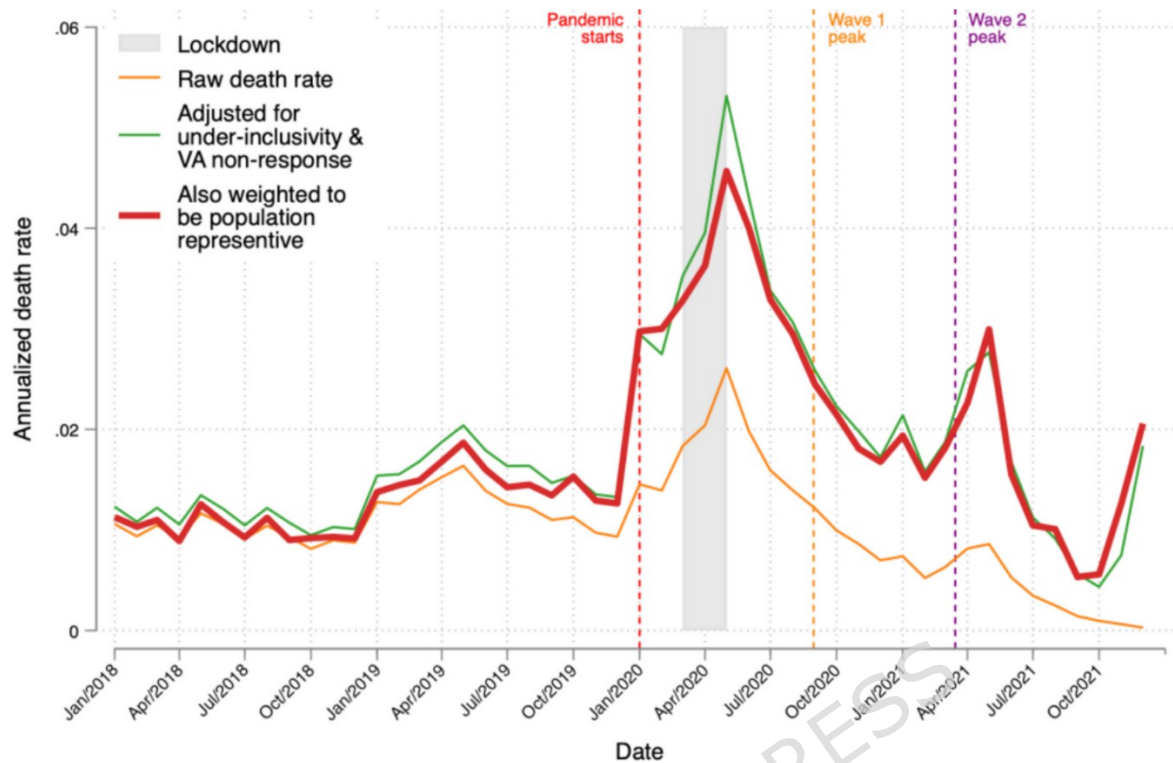


B: Excess deaths



Notes. Plot shows counts of deaths in the CPHS. These are equal to deaths recorded in the VA survey in the month of death, adjusted for non-response to the VA and the under-inclusivity of the CPHS.

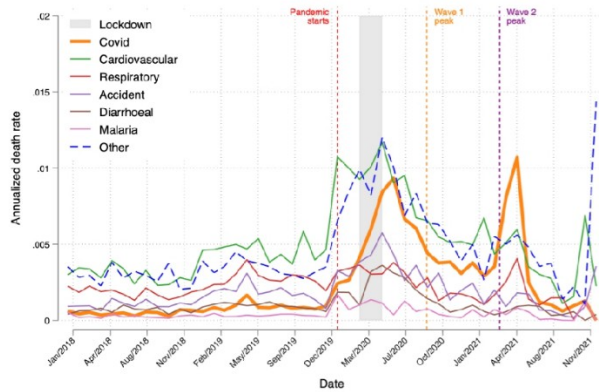
Figure 4. Monthly, all-cause death rate, 2018-2020.



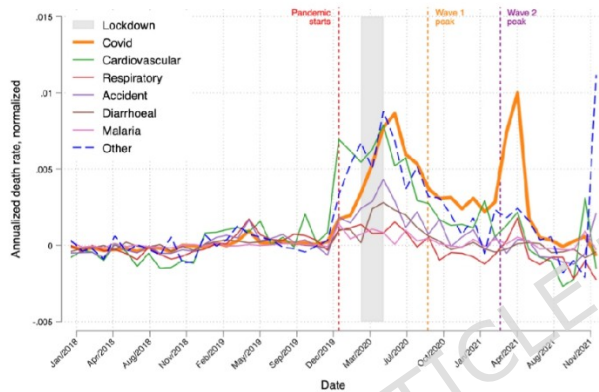
Notes. The raw death rate is the ratio of (a) the number of deaths in a month according to VA surveys and (b) the number of living persons in the CPHS data in that month. The adjusted death rate multiplies (a) by the adjustment factors for under-inclusiveness and for VA non-response. The weighted death rate also includes imputed CPHS sampling weights for deceased persons in the numerator and composite CPHS weights for living persons in the denominator.

Figure 5. Monthly population death rates by cause of death for SARS-CoV-2, top 5 non-SARS-CoV-2 causes of death, and other deaths, 2018-2021.

A: Absolute death rate

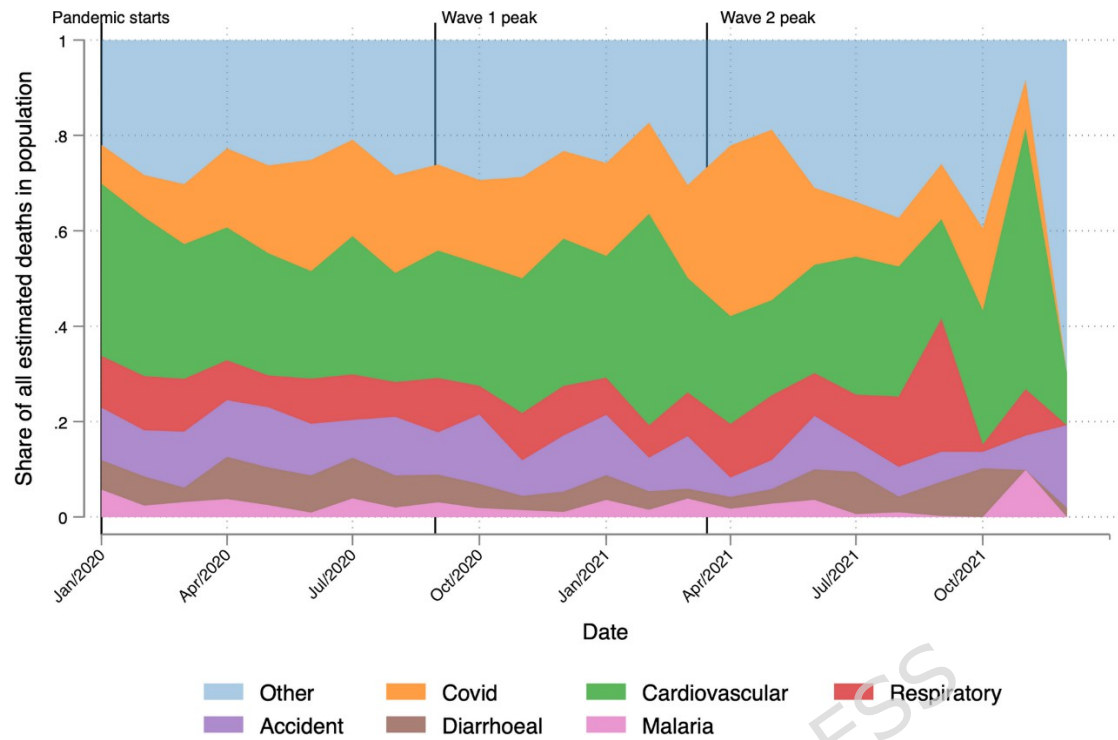


B: Index values



Notes. The population death rate in a month attributable to a cause is the fraction of people in the population that died from that cause in that month. It is estimated by the ratio of (a) the product of the non-response adjustment, the CPHS under-inclusivity adjustment, and the composite CPHS weight for each decedent summed over all decedents in the VA sample that month, and (b) the sum of CPHS composite weights for all living CPHS sample members that month.

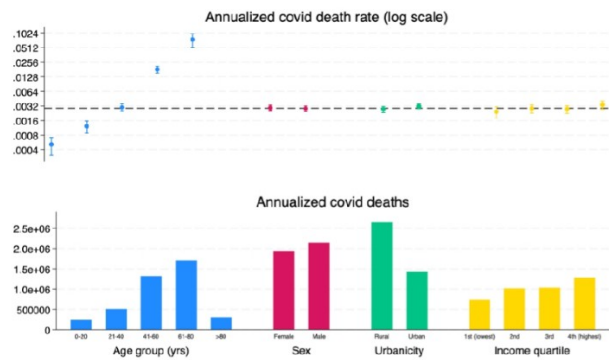
Figure 6. Share of death attributable to SARS-CoV-2 and 5 other top causes of death.



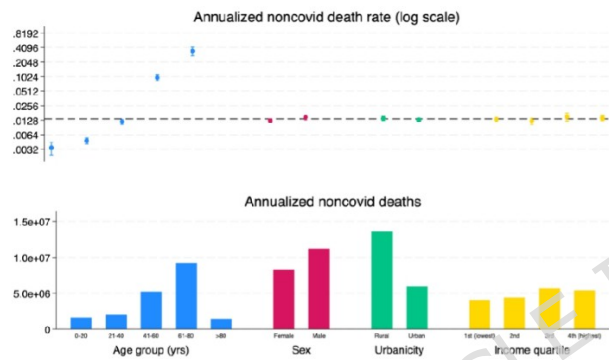
Notes. This figure shows the share of all deaths in each month attributable to the causes indicated in the legend.

Figure 7. Death rates and deaths from SARS-CoV-2 (panel A) and other causes during the pandemic, by age group and by rural or urban residence (panel B).

A: SARS-CoV-2

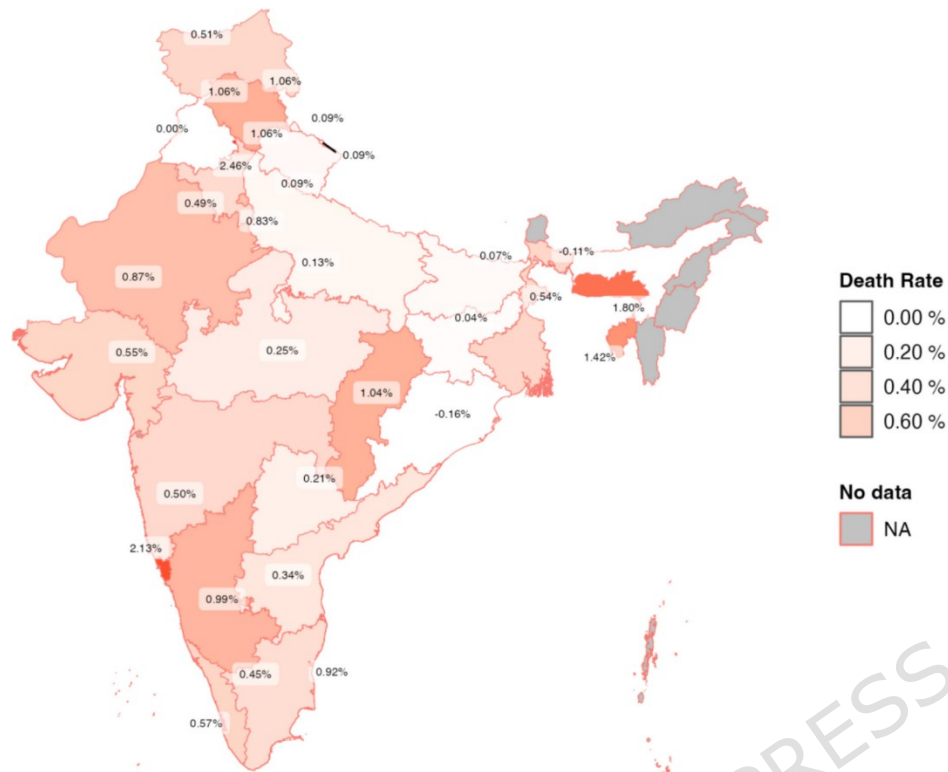


B: Other causes



Notes. Figures report estimated SARS-CoV-2 and non-COVID death rates and deaths in the population over time and by categories indicated in the panel titles. Horizontal dashed line gives the mean death rate for the relevant cause of, unconditional on age, sex, location, or income.

Figure 8. Excess SARS-CoV-2 death rates by state during the pandemic.



Notes. Plot presents the annualized, excess SARS-CoV-2 death rate for each state in 2020 and 2021. Excess death rates are defined as annualized U07 and U08 rates for 2020 and 2021 minus the average U07 and U08 death rates in 2018-2019.

Tables

Table 1. Cause-specific death and excess death rates.

Cause	Death rate		Excess death in 2020-21		Contribution to 2020-21 all cause	
	(1) 2018-19 (%)	(2) 2020-21 (%)	(3) (pp change)	(4) (% change)	(5) Death rate (%)	(6) Excess death rt (%)
All	1.25	2.26	1.01	81.19	100.00	100.00
COVID	0.07	0.41	0.34	484.06	17.93	33.15
Cardiovascular	0.38	0.61	0.23	61.75	27.18	23.15
Respiratory	0.23	0.21	-0.02	-6.83	9.32	-1.52
Tuberculosis	0.04	0.09	0.05	131.17	4.13	5.23
Neoplasm	0.06	0.07	0.01	12.03	3.20	0.77
Ill-defined	0.04	0.05	0.02	47.87	2.36	1.70
Digestive	0.05	0.09	0.03	63.07	3.85	3.32
Diarrheal	0.08	0.13	0.04	55.02	5.57	4.41
Accidents	0.14	0.23	0.09	62.18	10.22	8.74
Self-harm	0.02	0.03	0.02	79.70	1.53	1.51
Malaria	0.03	0.06	0.03	104.72	2.69	3.07
Hepatitis	0.04	0.05	0.02	41.44	2.27	1.49
Genitourinary	0.04	0.07	0.03	62.94	2.97	2.56
Unknown fever	0.02	0.03	0.01	67.40	1.13	1.02
Diabetes/endocrine	0.02	0.04	0.02	75.07	1.68	1.61
Epilepsy	0.01	0.02	0.01	104.83	0.76	0.87
Sexual diseases	0.00	0.00	0.00	-84.39	0.02	-0.18
Preventable by vaccine	0.00	0.00	0.00	138.57	0.11	0.15
Meningitis/encephalitis	0.01	0.01	0.00	28.66	0.33	0.17
Maternal	0.00	0.01	0.00	246.62	0.26	0.41
Other infectious diseases	0.03	0.04	0.01	31.71	1.94	1.04

Note. Columns 1 and 2 provide the death rate attributable to various causes and all causes in 2018-2019 (pre-pandemic) and 2020-2021. Columns 3 and 4 provide the excess cause-specific death rate in percentage points (pp) and percent of 2018-2019 same-cause death rate. Columns 5 and 6 provide the contribution each cause made to the 2020-21 all-cause death rate or excess death rate.